• **ELECTRICAL—ELECTRONIC SHIFT ON THE FLY (ESOF) TRANSFER CASE—INADVERTENT 4X4 HI OR LOW SHIFT EVENTS** • **TRANSMISSION—TRANSFER CASE—INADVERTENT 4X4 HI OR LOW SHIFT EVENTS—VEHICLES EQUIPPED WITH ELECTRONIC SHIFT ON THE FLY (ESOF) ONLY**	**Article No. 01-14-1**

FORD: 1999 F-250 LD
1999-2001 EXPEDITION, F-150

LINCOLN: 1999-2001 NAVIGATOR

ISSUE

Some vehicles may exhibit inadvertent 4X4 Hi or Low shift events, 4X4 and/or Low Range indicator flashing or solid or vehicle stuck in 4L after uncommanded shift. This may be caused by the 4X4 shift motor.

ACTION

Install a YL1Z-7G360-AA shift motor on vehicles built before 12/99. The YL1Z-7G360-AA shift motor with a grey contact plate cover replaces the prior design F75Z-7G360-AA motor with a blue contact plate cover. Grey cover shift motors contain improved sense plate material and shift motor terminal upgrade. For vehicles already equipped with the latest style shift motor, refer to the following Service Procedure for details.

SERVICE PROCEDURE

For overall description and operation of the Electronic Shift On The Fly (ESOF) Transfer Case, refer to Section 308-07 of the appropriate model year Workshop Manual.

NOTE
ALL DIAGNOSIS BY SYMPTOM SECTIONS HEREIN ARE WRITTEN ASSUMING THAT VEHICLE IS ALREADY EQUIPPED WITH THE LATEST STYLE SHIFT MOTOR. ALWAYS REPLACE SHIFT MOTOR AS A FIRST STEP IF CURRENT MOTOR IS PRIOR LEVEL WITH BLUE COVER.

NOTE
TO RESET THE GEM'S 4X4 MONITORING SOFTWARE, USE CLEAR DTC COMMAND IN THE NGS OR WDS TESTER (EVEN IF NO DTC'S ARE AVAILABLE) AND CYCLE IGNITION KEY ON-OFF-ON. RECORD ANY CONTINUOUS MEMORY CODES PRIOR TO CLEARING DTCs.

NOTE
AN UNCOMMANDED 4H OR 4L SHIFT DUE TO FALSE SHIFT MOTOR CONTACT PLATE SIGNAL MAY OR MAY NOT BE ACCOMPANIED BY SHIFT MOTOR DTC'S P1838, P1846, P1850, P1854, P1858, P1866, AND/OR P1867 STORED IN THE GEM DEPENDING ON WHETHER THE IGNITION KEY WAS CYCLED AFTER THE EVENT.

NOTE
CONTACT PLATE PIDS READ OPPOSITE OF 1999 OR EARLIER WORKSHOP MANUALS WHEN VIEWED WITH NGS RED CARD VERSIONS 15.0 OR LATER. THIS TSB CONTENT REFLECTS USING NGS VERSION 18.0.

UNCOMMANDED SHIFT TO 4L

1. **Parking Maneuvers/Gear Lever Transitions:** Symptoms noted during a lower speed uncommanded shift to 4L event include:
   - Front end binding or hopping while turning
   - Bind feel in drivetrain when backing up and/or turning
   - Audible clunking or grinding noises, and/or
   - Amber low range light illuminated

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a. If the Generic Electronic Module (GEM) receives a false Mode Switch input during or shortly after the Digital Transmission Range Sensor (DTR) indicates a Neutral range pass-thru, an uncommanded shift is possible if remaining 4L pre-conditions are met. The pre-conditions besides transmission in Neutral range include: service brake depressed, and vehicle speed less than 5 km/h (3 mph). A false switch input may also set a P1812 or P1815 DTC in the GEM.

b. Check mode switch circuits 682 (dark blue) for short to power and 780 (dark blue) for ground short, loose connections at inline connectors, and chafes.

c. Monitor Mode Switch Parameter Identification Display (PID) 4WD_SW for false readings while slightly pushing in and wiggling the mode switch knob. Refer to Workshop Manual Section 308-07A for mode switch resistance tests for all switch positions.

d. Replace Mode Switch if it fails testing or repair wiring on circuits 682 or 780 if fault indicated. If tests pass, inspect build date stamped on GEM and replace if GEM built prior to 9/98.

e. Test drive vehicle for proper 4X4 operation in all modes. Include parking lot maneuvers and transmission gear range lever transitions while testing. Pull DTCs from GEM (even if there are no DTCs available). Clear any codes present and cycle ignition.

2. **Road Speed:** Reported symptoms for a higher speed uncommanded shift to 4L event include:
   - Rapid deceleration
   - High engine revving possibly accompanied by a P1270 DTC in the PCM
   - Clunk/grind noise
   - Speedometer may spike higher than actual speed.

After the initial event the vehicle may exhibit restricted vehicle top speed without 4X4 indicators illuminated but possible MIL (Malfunction Indicator Lamp) on.

a. Monitor contact plate A,B,C,D PIDS. With Mode Switch in;
   - A4WD, plate PIDS should read "OCOO"
   - 4H PIDS should read "COOC"
   - 4L PIDS should read "COCO" (sequentially read starting with plate A then B, then C, finally D).

If contact plate PIDS DO NOT correspond to the set 4WD position, check for continuity/shorts/moisture/corrosion in the vehicle side of the transfer case shift motor connector (toward the GEM). Visually inspect all terminals, pins, crimps, and connectors closely. Repair any wiring conditions in the contact plate circuits as necessary.

b. Road test at speeds above 16 km/h (10 mph) with Mode Switch in both A4WD and 4H to see if the condition returns. Clear DTCs from GEM (even if no DTCs exist) and cycle ignition. Clear DTCs from PCM if a P1270 (engine RPM or vehicle speed limiter reached) code was initially present.

UNCOMMANDED SHIFT TO 4H

1. **Autolock Strategy (Expedition/Navigator Vehicles Only):** Uncommanded 4H shifts with red 4X4 light on "solid" while driving in Automatic Mode may be a vehicle characteristic if driven off-road or under slippery conditions. The GEM Autolock strategy commands transfer case clutch to minimum duty cycle and engages 4H (with mode switch still set to A4WD) to prevent continuous cycling from prematurely wearing the clutch. This is design intent. Excessive tire circumference variations or axles with unmatched gear ratio's can result in unexpected Autolock function as well, for more detail check the speed sensor section below. Once the condition that caused Autolock strategy to activate is no longer present, the system stays in 4H until operator cycles the ignition key or sets the mode switch to 4H then back to A4WD.

2. **Transfer Case Speed Sensors (Expedition/Navigator Vehicles Only):** If uncommanded shift to 4H occurs on hi-traction surfaces, check TRA_FSP and TRA_RSP PIDS for excessive speed sensor variation between the front and rear transfer case speed sensors.

a. If PIDS show sensor inputs not within 2-3 km/h (1-2 mph) of each other during steady-state driving with possible DTCs P1836/P1837, first verify tire pressures, sizes, circumference within 1/4 ″ among all four tires, and front and rear axle gear ratio's for matching.

NOTE
DISCONNECTING CIRCUIT 779 (BROWN) AT MAGNETIC CLUTCH COIL ALLOWS MONITORING SPEED SENSORS WITHOUT CLUTCH ENGAGEMENT INTERFERENCE, AND WILL NOT SET ANY DTCs OR AFFECT DRIVEABILITY. BE SURE TO RECONNECT CIRCUIT WHEN TESTING IS COMPLETED.

b. If ratio's match and tire sizes are okay, check associated circuits and replace speed sensor(s) or repair wiring as necessary.

c. During sensor installation use lubricant on sensor O-ring and verify fully seated before tightening hold-down bracket.

d. After any repairs, road test while comparing NGS GEM PIDS TRA_FSP and TRA_RSP to PCM VSS signal. Verify all three PIDS match each other within 2-3 km/h (1-2 mph).

3. False Input to GEM A false mode switch input on circuits 682 and/or 780 (Dark Blue) could be interpreted by the GEM as operator requesting a 4H shift. Under this condition, the GEM will command shift motor movement to match the false switch input. The red 4X4 indicator will illuminate just as if operator moved the switch. If the condition became intermittent, the 4X4 range and also the indicators could switch back and forth between A4WD/2H and 4H.

   a. Check the Mode Switch circuits between the GEM and dash for chafing, shorts, crimps, continuity. Repair wiring if faults found or if wiring okay, replace the Mode Switch itself.

   b. After any repair, pull DTCs from the GEM (even if there are no DTCs available). Clear any codes present and cycle ignition.

STUCK IN 4L AT ALL TIMES

1. If the customer describes an uncommanded shift to 4L shortly after start-up and then unable to shift transfer case, check operation. Associated symptoms could include valid GEM DTCs being erased and false DTCs such as C1107 present. If stuck in 4X4, re-establishing normal shift motor operation is possible by putting mode switch in 4L with trans gear lever in neutral, service brake depressed, and vehicle stationary. To address:

   a. Replace the GEM module.

   b. Test drive vehicle. Include a start-up after soak period trying to match the conditions initially reported by the customer to verify the repair.

   c. Pull DTCs from GEM (even if there are no DTCs available). Clear any codes present and cycle ignition.

PART NUMBER	PART NAME
YL1Z-7G360-AA	Shift Motor - Transfer Case (ESOF)

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 206000, 508000, 597997, 610000, 610600, 610700, 612000, 612600, 698298